

Table 1: Primary physical-complementarity summaries. The reported combined-set gains are the archived grid-wide improvement statistics relative to ringdown, not ratios formed from separately reported medians.

Features	$\text{med}(\sigma_{\min})$	$\text{med}(\kappa(J))$	gain
Ringdown	0.2782	6.6112	reference
Ringdown + photon geometry	1.2326	4.0731	$4.54 \times / 2.28 \times$
Ringdown + all shooting	1.5312	4.4974	—

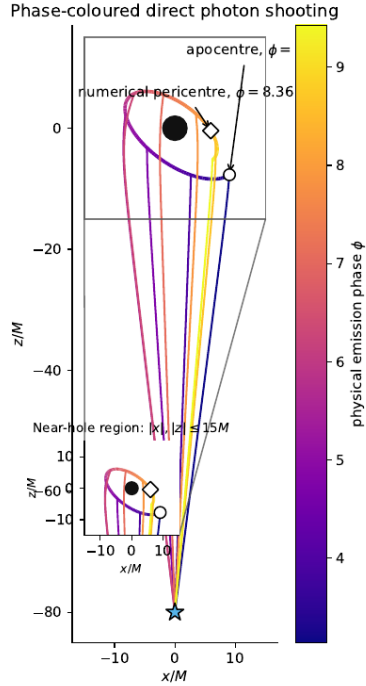


Figure 2: Representative direct-branch null geodesics for $M = 1$, $r_p = 8M$, $r_a = 12M$, $k = 10^{-3}$, $w_q = -0.5$, and observer $(0, 0, -80M)$. Physical phase runs from π at apocentre to 3π ; the diamond marks the numerically detected pericentre and the dark disk is a horizon-scale marker, not a shadow. At each phase, two launch angles are solved until the ray reaches the observer. The inset enlarges the stated near-hole coordinate region without changing the trajectory scale. This is a controlled dimensionless benchmark.

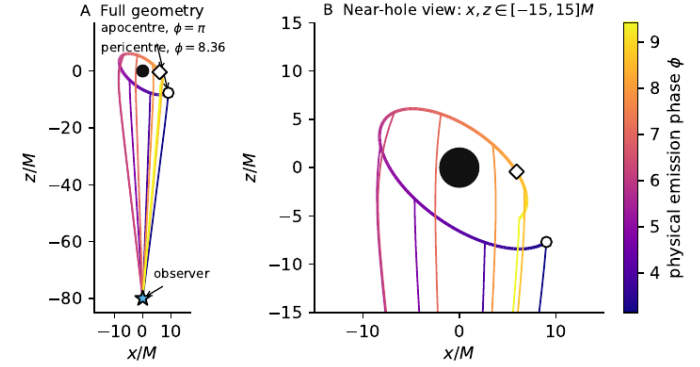


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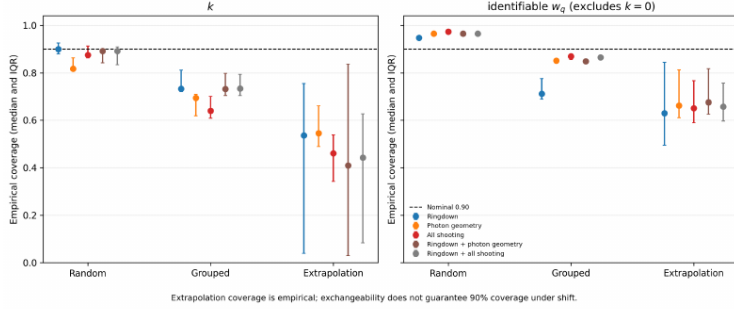


Figure 10: Empirical 90% split-conformal coverage by target and protocol. Ordinary w_q coverage excludes $k = 0$; extrapolation coverage is not guaranteed under distribution shift.

Model	median	95th percentile	max. identifiable w_q
HGB	0	0.08122	0.30593
RF	0	0.08509	0.27053
MLP	0.001143	0.008223	0.03819

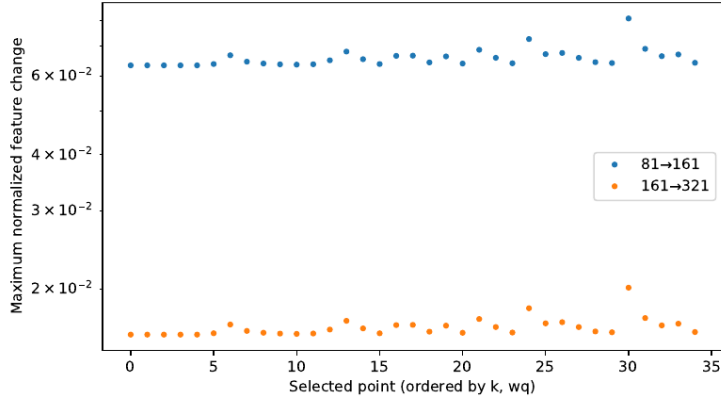


Figure 11: Targeted pointwise maximum normalized feature changes from 81 to 161 and 161 to 321 phases. The 321-phase audit resolves the forward-feature criterion but does not by itself certify inverse estimators.

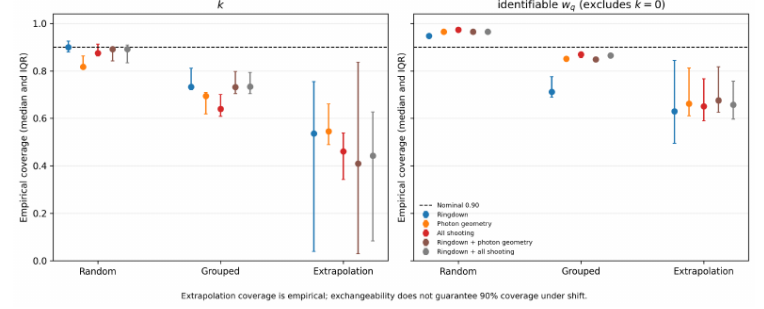


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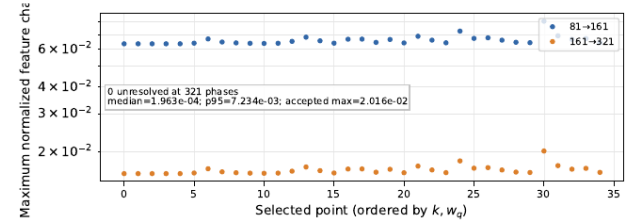


Figure 11: Targeted pointwise maximum normalized feature changes from 81 to 161 and 161 to 321 phases. The 321-phase audit resolves the forward-feature criterion but does not by itself certify inverse estimators.

tary observables, partition by physical systems, test directional shift, and perturb frozen estimators at independently measured numerical scales.

9 Limitations

The ringdown model is leading-eikonal rather than a dedicated finite- ℓ perturbative spectrum. The spacetime is static. Shooting includes only the direct photon branch, a point emitter and point observer, and no radiative transfer, detector response, secondary images, strain likelihood, calibration error, or model discrepancy. The physical grid is finite. Historical proxy results remain synthetic. Although the targeted audit resolves higher-harmonic forward differences on its selected set, it is not a uniform 321-phase grid and tree tail sensitivity persists. MLP extrapolation is unstable. The verdict is therefore conditional, and no result is an observational constraint.

10 Conclusion

Physical complementarity is supported: ringdown constrains k , while photon geometry supplies an independent direction needed for much of the w_q recovery. Random interpolation overstates recoverability relative to grouped and directional validation. Finally, numerical and distribution-shift audits are necessary before confident inverse claims: the forward features can converge while fitted estimators remain fragile. These conclusions motivate robust inverse methods and broader physical modeling before observational use.

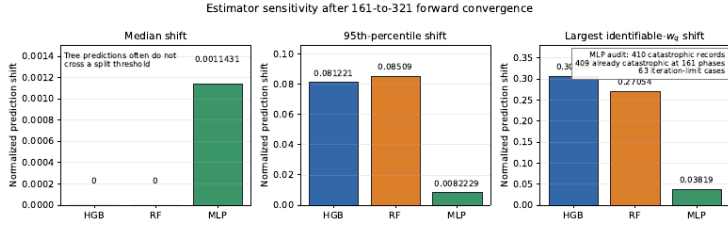


Figure 12: Frozen-estimator sensitivity after forward convergence. Bars show the median, 95th-percentile, and largest identifiable- w_q normalized prediction shift for HGB, RF, and MLP. The annotation records the distinct MLP failure audit: 410 catastrophic directional records, 409 already catastrophic at 161 phases, and 63 iteration-limit cases. No predictions are clipped.

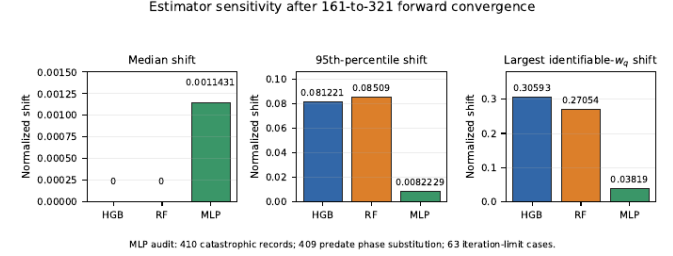


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A Historical Synthetic and Proxy Benchmark

The original dense benchmark used leading-eikonal curves, training-fold PCA, and synthetic geodesic proxies. Waveform-only RF achieved $R^2(k) = 0.8960$ and $R^2(w_q) = 0.9587$; HGB with proxy augmentation achieved 0.9793 and 0.9891. The rank-aware classifier reached macro F1 0.9928 with a 57.6% weak-identifiability fraction. PCA compressed correlated information; it did not create a new physical response direction. These historical results motivated the physical shooting study and are not substituted for it.

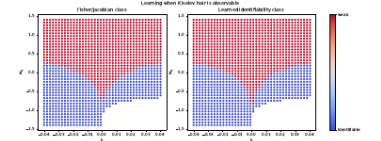


Figure 13: Historical dense synthetic rank-aware identifiability boundary. Exact $k = 0$ rank loss is structural, whereas the finite-width weak-identifiability region depends on precision. This historical benchmark motivated, but does not replace, the validated physical shooting analysis.

B Full Split Definitions

Random splits partition parameter points before model-level realization. Grouped physical interpolation holds out blocks under both registered 3×3 layouts. Directional extrapolation separately holds high k , low k , high w_q , and low w_q sides. Calibration points are disjoint from training and test points, and the archived assignments are reused exactly.

C Additional Model-Level Results

HGB and RF reproduce the central grouped photon-geometry improvement for identifiable w_q . The MLP is retained as a stress test rather than omitted: its catastrophic errors are un-clipped and included in aggregate summaries.

D Four Directional Extrapolation Tests

Directions are stored independently in the prediction tables before any median summary. High-to-low and low-to-high tests differ in training support and cannot be interpreted as interchangeable folds.

E Conformal Calibration and Rejection

Split-conformal residual quantiles are fitted on calibration subsets. The exchangeability guarantee applies to matching distributions; shifted coverage is reported empirically. Rejection curves rank uncertainty but do not repair distribution shift.